

UCE618: SUSTAINABLE CONSTRUCTION PRACTICES				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: The objective of this course is to expose the students to the concepts of sustainability in the context of building and conventional engineered building materials, such as Concrete, Bricks, and achieving the same through lower Carbon cements, Superior brick kilns and Recycled aggregate minimizing consumption of natural resources including water.				
Principles of Sustainability: Basics of Carbon Cycle, Fundamentals of Sustainability, Triple Bottom Line, Embodied carbon and embodied energy				
Sustainable Materials: Role of Materials: Carbon from Cement, alternative cements and cementitious material, Alternative fuel for cements for reduction in carbon emission, Sustainability issues for concrete, Types of Composite Cement, Natural Building Materials and systems; Clay bricks and Types of Kilns				
Sustainable Construction: Role of quality control, minimization of natural resource utilization, geo-polymer concrete, concrete with alternative material for sustainability, Curing Methods and Use of Waste Water for Mixing and Curing; Sustainable Waste Management; Sustainable construction methods like low carbon construction, off site manufacture, etc. Introduction to design for Durability method for sustainable mix design				
Green Buildings: Operational energy reduction and net zero building, Optimization for design of building for energy efficiency, Thermal Diffusivity, Green Performance rating, requirements of LEED, GRIHA etc.				
Exercise/ Assignments Case studies related to the application of traditional and advanced technology for sustainable construction.				
Course Learning Objectives (CLO) Upon completion of this course, the students will be able to: 1. Understand importance of sustainability in construction 2. Develop concrete with alternative sustainable Construction materials and methods 3. Get familiar with international mix design based on durability 4. Understand the definition and concept green buildings				
Text Books 1. Concrete Technology by A. M. Neville, J.J. Brooks, 2010 2. Sustainability of Construction Materials, Jamal Khatib, 2016				
Reference Books 1. Siddique R, Cachim P. Waste and Supplementary Cementitious Materials in Concrete: Characterization, Properties and Applications. Woodhead Publishing 2. Michael Thomas, Supplementary Cementing Material in Concrete, CRC Press, 2013 3. Satish Chandra, Waste materials used in concrete, Noyes Publications, 1997 4. BS 8500-1 (2015): Concrete–Complementary British Standard to BS EN 206-1–Part 1: Method of Specifying and Guidance for the Specifier				

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	25